

Federico Melo Barrero

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EDUCATION

Universidad de los Andes

Bogotá, Colombia | 2020 – 2024

B.Sc. Systems and Computing Engineering

- GPA: 4.92/5.00
- *Summa cum laude*, highest GPA in major in 40 years, highest in Engineering Faculty in 16 years
- Recipient of the Quiero Estudiar scholarship (95% tuition coverage) for the duration of the degree

AWARDS & HONORS

Dejar Huella Award

Universidad de los Andes, 2026

Universidad de los Andes recognition for team impact on the university community. Awarded for "No Estás Solo," a student-built peer-support platform serving 56,000+ active students with ~1,000 daily connections to institutional support

Top Saber Pro Score Award 2024

Colombian Ministry of Education, 2025

Awarded to top-performing students on Colombia's official national standardized higher-education examinations (Saber Pro and TyT)

Summa Cum Laude

Universidad de los Andes, 2025

Summa Cum Laude is the highest undergraduate academic honor. Eligibility requires a cumulative GPA within the top 1% of graduates from the faculty over the past five years, and demonstrated integral merits in an interview before a commission appointed by the Academic Council

Athletic Career Distinction

Universidad de los Andes, 2025

The Athletic Career Distinction is awarded to a single student in the graduating class for exemplary leadership, commitment, and ethics in sports. Awarded citing 41 swimming medals, including 24 gold, and two years as elected team captain

Ramón de Zubiría Award (2 times)

Universidad de los Andes; 2023, 2024

Annual distinction awarded to the undergraduate student with the highest cumulative GPA in their major

Semester Excellence Award (3 times)

Universidad de los Andes; Spring 2021, Fall 2021, Fall 2023

Awarded to the undergraduate student with the highest semester GPA in their academic program during the preceding semester

Quiero Estudiar Scholarship

Universidad de los Andes, 2020

Merit- and need-based scholarship from Universidad de los Andes covering 95% of tuition for the full undergraduate degree

WORK EXPERIENCE

Canals AI

Coral Gables, FL, USA (Remote) | Dec 2024 – Present

Software Engineer II

Jul 2025 – Present

- Owned parsing services for the live voice and inquiries agent systems, as part of a pipeline that processed 20000+ orders daily across 115+ wholesale distributors and 9000+ sales reps
- Built the parsing and review layers of a real-time voice service (WebSocket orchestration, speech-to-text, speaker diarization, concurrency control). Extracted ~21 products per call from live phone calls with 75%+ retained through rep review
- Integrated tools (web search, spec sheet retrieval, ERP lookup, order tracking) into an inquiries AI agent that addressed customer questions on orders, materials, and logistics. Achieved 97% of answers citing a source and 80% rated positive by reps
- Contributed to Canals' shared agent framework (Mastra), adding extended thinking, reasoning effort, image/PDF input, temperature, and tool concurrency. Migrated four Parsing agents off hand-rolled, unbounded loops onto bounded runs with step and token ceilings
- Co-built an autonomous agent (Claude Agent SDK) that polled tickets, debugged parsing failures, and shipped prompt-level fixes for review, re-parsing each order to verify its own fix
- One of 4 engineers on the 10-person Parsing team (~60 engineers total) holding merge approval rights on the main branch.

Software Engineer I

Dec 2024 – Jul 2025

- Maintained a parsing pipeline supporting 19 mimetypes with intricate formats (e.g., handwritten, struck-through text, blurry and fragmented photos, cross-references to other files) in a pipeline serving ~60 wholesale distributors processing 4,000+ orders daily
- Abstracted OCR pipeline from single provider to support multiple providers, decoupling the implementation from Textract and enabling format-specific OCR selection, with a cache layer serving ~60% of requests in 0.07s versus 3.6–18.7s uncached
- Led large-scale prompt refactors improving external QA pass rate from 85% to 89%. Involved: model selection, prompt engineering, cache handling, fine-tuning, and regression testing
- Built visual highlighting systems using DOM traversal, coordinate mapping, and string comparison to map parsed products to source documents. Reduced weekly highlighting errors by 2/3
- Joined as the company's 14th engineer.

Universidad de los Andes

Bogotá, Colombia (Part-time) | Jan 2024 – Dec 2024

Full-Stack Developer

- Designed and built full-stack student data analytics system serving 15,000+ students and 600+ faculty, replacing an external platform and saving ~\$100K USD annually
- Built parallelized data processing pipeline (Python, Polars) with complex filtering and aggregations, exposed through a unified REST API (FastAPI)
- Developed dynamic graphing, advanced filtering, report generation, and role-based dashboards in a React/TypeScript web app
- Built early-warning system for at-risk students (academic performance, dropout indicators), alerting counselors on 1,000+ critical cases
- Architected secure cross-VM communication using air-gapping and network isolation techniques to ensure compliance with student data protection requirements

Caseware International Inc.

Toronto, ON, Canada (Remote, Internship) | Jan 2024 – Jul 2024

Software Developer I

- Awarded the June Team Award for 'delivering impactful contributions and hard work' as part of the Data Analytics team
- Developed Python-based Jupyter notebooks for financial and auditing data analysis
- Optimized Java and TypeScript microservices performance using AWS S3 and DynamoDB, identifying and resolving bugs
- Authored and delivered technical reports (SPIKEs, RFCs) that influenced team architecture decisions and established best practices

Universidad de los Andes

Bogotá, Colombia (Hybrid, Part-time) | Aug 2023 – Dec 2023

Backend Developer

- Designed and implemented backend architecture for the Professors' Portfolio system (Java, Spring Boot), a web application for managing faculty academic and administrative information.
- Built persistence, logic, and control layers with CRUD operations, custom logic, and RESTful APIs
- Devised efficient data migration methods to store large volumes of unstructured faculty data into relational databases
- Collaborated with Prof. José Bocanegra, Ph.D. on system architecture documentation and technical specifications, contributing to project planning and implementation roadmaps

TEACHING EXPERIENCE

Universidad de los Andes

Bogotá, Colombia | Aug 2025 – Present

Adjunct Professor, Algorithm Design

Fall 2026

- Principal instructor for upper-division algorithm design course, delivering lectures and designing exams
- Topics: divide and conquer, dynamic programming, graph algorithms, NP-completeness, approximation and randomized algorithms

Adjunct Professor, Introduction to Programming

Fall 2025, Spring 2026

- Principal instructor for introductory undergraduate course in procedural programming using Python
- Topics: programming fundamentals and data-processing libraries Pandas and Matplotlib

Universidad de los Andes

Bogotá, Colombia | Summer 2024

Undergraduate Teaching Assistant, Vector Calculus

Supervisor: Prof. Jacinto Puig, Ph.D.

- Designed and graded problem sets, and conducted weekly office hours
- Covered partial derivatives, double and triple integrals, line and surface integrals, vector fields, curl and divergence calculations, and the fundamental theorems (Green's, Stokes', and Gauss')
- Helped students apply these concepts to real-world problems, mainly involving constrained and unconstrained optimization

Universidad de los Andes

Bogotá, Colombia | Fall 2022 – Fall 2024

Undergraduate Research Teaching Assistant, CupiTaller

Supervisor: Prof. Eduardo Rosales, Ph.D.

- Collaborated in a 7-person team under Prof. Eduardo Rosales, Ph.D., to manage the university's programming support center
- Oversaw the university's Coursera courses 'Programming in Python' and 'Introduction to Object Oriented Programming in Java', which have 50,000+ and 60,000+ historical enrollments respectively
- Managed schedules for ~40 tutors and ~70 TAs each semester, overseeing a system that enabled 1,500+ students to book tutoring sessions
- Designed and co-built CupiHorarios, a scheduling web app (FastAPI, React) leveraging Pyomo (a constraint satisfaction optimization solver) to generate schedules that ensured full coverage during operational hours
- Co-designed and co-built CupiFeedback, a tutoring performance analytics web app (FastAPI, React) that collected feedback from 1,500+ students and presented insights through interactive visualizations
- Co-designed CupiMonitores, a web app that centralized TA evaluation, which was ultimately adopted university-wide
- Developed automation systems including a TypeScript Discord bot for remote tutoring tracking and programming problems with automated solution verification
- Led tutor recruitment and training processes, conducting candidate interviews and occasionally providing coverage during tutor absences, teaching data structures and algorithms concepts

RELEVANT COURSEWORK

Computer Science: Languages and Automata, Reinforcement Learning, Modeling, Simulation and Optimization, Algorithm Design and Analysis

Mathematics: Graph Theory, Game Theory, Computational Methods

Software Engineering: Software Metrics and Quality, Software Architecture and Design, API Design and Construction

PUBLICATIONS

Martínez Novoa, S., **Melo Barrero, F.**, Ruiz Giraldo, M., Velásquez Marín, S., Carvajal Chaves, N., Hernández Hoyos, M., Reyes, J. P. (2026). "Centralizing the Student Lens: Design and Evaluation of a Learning Analytics Dashboard for Academic Advising in Higher Education". *Proceedings of the World Engineering Education Forum (WEEF 2026)*. IFEEES & GEDC. To appear.

Melo Barrero, F. (2024). "Desarrollo del Perfil del estudiante dentro de No estás solo". *Undergraduate thesis*. Universidad de los Andes. Available at: [Uniandes Repository](#)

TECHNICAL EXPERIENCE

AI Engineering & LLM Orchestration: Anthropic / OpenAI / Gemini / Groq APIs, Claude Agent SDK, Mastra, MCP

Backend & Systems Architecture: Python, Java, TypeScript, FastAPI, Spring Boot, WebSocket
Orchestration, Concurrency Control

Data Engineering & Infrastructure: Polars, PostgreSQL, Redis, AWS (RDS, ECS, S3, DynamoDB),
Docker, GitHub CI/CD

Product & Interface: React, Next.js, Svelte, Plotly, ShadCN / MUI

EXTRACURRICULAR ACTIVITIES

National-level swimmer

- 1st place, men's 20–29, 2 km open water, OCEANMAN San Andrés 2026
- Member and captain of the Universidad de los Andes Swimming Team, earning 24 gold, 10 silver, and 7 bronze medals
- Four-time member of the Bogotá City Swimming Team, competing in 12 National Swimming Championships

ADDITIONAL INFORMATION

Languages: Spanish (native), English (IELTS Academic 8.5 / CEFR C2)